

## 02 Soc2

### SOC 2 Type II Readiness — L\_LIGHTRAG

**Company:** Anticloud FZ LLE **Model:** PAX L5 Narrow L2 General 27B

**Trust Service Criteria:** Security, Availability, Confidentiality, Processing Integrity, Privacy

#### CC1 — Control Environment

Criteria Evidence Status

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CC1.1 COSO Principles Documented in OFFICIAL\_BENCHMARKS PASS

CC1.2 Board oversight Anticloud FZ LLE governance docs PASS

CC1.3 Management philosophy Single-owner accountability (Lois-Kleinner Alpasan) PASS

#### CC2 — Communication & Information

Criteria Evidence Status

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CC2.1 COSO objectives Anticloud offline-first architecture documented PASS

CC2.2 Internal communication Local observability stack (traces/metrics/logs) PASS

CC2.3 External communication Anticommons 0.1.0 public license PASS

#### CC6 — Logical & Physical Access (Security)

Criteria Evidence Status

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CC6.1 Logical access security Zero-trust mTLS mesh, short-lived certs PASS

CC6.2 Authentication mTLS client certificates, no shared secrets PASS

CC6.3 Role-based access CRDT identity + AIOSS audit per role PASS

CC6.6 Logical access restrictions Air-gapped deployment option PASS

CC6.7 Unauthorized removal AES-256-GCM encryption, key rotation PASS

CC6.8 Malware prevention Deterministic builds + source hash verification PASS

#### CC7 — System Operations

Criteria Evidence Status

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CC7.1 Configuration management Deterministic builds, SBOM in binary PASS

CC7.2 Infrastructure monitoring Local observability: OpenTelemetry traces PASS

CC7.3 Environmental threats Offline-first, no cloud dependency PASS

CC7.4 Incident detection AIOSS anomaly detection via chain breaks PASS

#### CC8 — Change Management

Criteria Evidence Status

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CC8.1 Change management Provenance chain records every computation PASS

## **A1 — Availability**

99.99% theoretical uptime — no cloud dependency means no cloud outage risk.  
Single binary deploys in < 30 seconds on any x86/ARM node.

## **C1 — Confidentiality**

AES-256-GCM encryption at rest. Zero plaintext storage of sensitive data.  
Sovereign Memory encrypts all persisted entries with script-derived keys.

## **PI1 — Processing Integrity**

AIOSS ledger provides cryptographic proof of processing integrity.  
Every computation recorded in Provenance chain with input/output hashes.